

- 31** Laser light of a single frequency is incident normally on a diffraction grating with $5.00 \times 10^5 \text{ lines m}^{-1}$. A diffraction pattern is formed on a flat screen which is parallel to the grating.

The screen is a distance of 33.5 cm from the grating. The distance between the two first-order intensity maxima is 15.0 cm.

What is the wavelength of the light?

- A** 437 nm **B** 448 nm **C** 817 nm **D** 1950 nm